

□ ALLUVIAL SOIL — FULL MIND MAP

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📖 Topics in this chapter:

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2. Composition — Potash/Phosphoric Rich, Nitrogen Poor
3. River-Deposited Origin — Khadar vs Bangar
4. Sand Content Gradient & Area Coverage (40%)
5. Bhangar — Old Kachhari Clay
6. Water Retention — Sandy vs Clayey/Loamy
7. Loamy Soil Composition
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1. Alluvial Soil — India's Largest & Most Fertile Group

- Alluvial soil India ka **sabse bada soil group hai** — ~15 lakh sq km (45.6% total area), most **productive/fertile soil** — Red, Black, Calcareous soil se zyada productive. [Q1, Q2, Q5, Q6]

2. Composition — Potash/Phosphoric Rich, Nitrogen Poor

- Alluvial soil **Potash, Phosphoric acid, Alkalines mein rich hai** — **Nitrogen ki proportion kam hai**. Saline soils ke local naam: **Reh, Thur, Chopan, Kallar, Rakar, Karl** (in saline hone ka matlab yeh nahi ki poori Alluvial soil infertile hai — overall Alluvial soil bahut fertile hai). [Q1, Q4]

3. River-Deposited Origin — Khadar vs Bangar

- Alluvial soil **rivers ke silt se bani hai (Himalayan sediments + sea-retreat)** — do sub-categories: **Khadar (new alluvium, floodplain mein, har saal naya deposit) aur Bangar (ancient alluvium, floods ki reach se upar)**. [Q3]

4. Sand Content Gradient & Area Coverage (40%)

- Alluvial soil ke sahi facts: **sand content west se east ki taraf kam hoti jaati hai, Bhangar-Khadar forms mein hai, total area ka ~40% cover karti hai (NCERT ke mutabik), potash mein rich hai lekin phosphorus mein poor hai** — sab 4 statements sahi hain. [Q4]

5. Bhangar — Old Kachhari Clay

- Gangetic plain ki purani Kachhari clay ko '**Bhangar**' kehte hain — **floods ki reach se pare, higher level par milta hai**, calcareous concretions hote hain, pale reddish-brown color ka hota hai. [Q7]

6. Water Retention — Sandy vs Clayey/Loamy

- Water retention capacity comparison mein **Alluvial sand soil ki sabse kam water retention capacity** hoti hai — loamy sand, clayey loam, loamy soil se kam. [Q8]

7. Loamy Soil Composition

- Loamy soil mein **sab types ke particles** (~40% sand, 18% clay, 42% silt) hote hain. *[Q9]*

| 8. Chhattisgarh's Local Soil Names — Bhatha's Low pH

- Chhattisgarh ke local soil naam: **Bhatha** (laterite type, lowest pH ~6.5, acidic), **Dorsa**, **Matasi**, **Kanhar** — Bhatha soil **leached hai**, **sandy-loam** se **gravelly-sand texture ki hai**, **red/red-yellow color ki hai**, often barren land classify hoti hai. *[Q10]*